CHAPTER 1
INTRODUCTION

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Chapter Objectives

Upon completion of this chapter, the student will be able to:

- Define terms used & classify computers
- Distinguish between HW & SW
- Briefly describe the various HW parts
- Understand the various roles of SW

Sub-topics

1.0 Definitions & classifications
1.1 Introduction to I/O devices
1.2 Introduction to CPU
1.3 Storage devices
1.4 The expansion cards
1.5 The electrical system
1.6 The 3 types of software

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1.0 Definitions & classifications

Definition of a computer

Electronic device that accepts data (I/P), processes it into information, stores information & outputs it (O/P).

Data means raw facts & information means processed data

Classification of computers by size

1. Mainframe
2. Miniframe
3. Microcomputer

Mainframe

- Large computers owned by big companies
- High speed & multi-user

Supercomputers
- Also nicknamed *monsters*, are the biggest mainframes
- Used for projects e.g. weather forecasting, nuclear weapon projects

Miniframe
- Smaller than mainframe, owned by big Companies.
- Multi-user, less users than mainframe

Microcomputer
- Much smaller than & less expensive than the above
- Single user & so commonly called PC (Personal Computer)
- Appeared in 1970s

Types of PCs
1. Desktop PCs
   Monitor rests on top of the system unit (box)
2. Tower PCs
   System unit (box) stands upright beside the monitor, to save on office space
3. Laptop
   Much smaller PCs that can be carried in a bag or briefcase
   Notebooks & palmtops are much smaller laptops

Brands of PCs
Include:
Compaq, Dell, HP, IBM, Acer, iNTex, Toshiba, Fujitsu

Brief history of PCs
1971-1st PC sold to an individual with a TV screen
1975-1st PC made with a screen & a keyboard, sold in a kit form
1981-1st PC made by IBM (International Business Machines)
1983-1st PC made called a Macintosh (MAC) by Apple with a mouse graphics
Note: IBM & Apple have been competitors in manufacture of PCs

Computer system

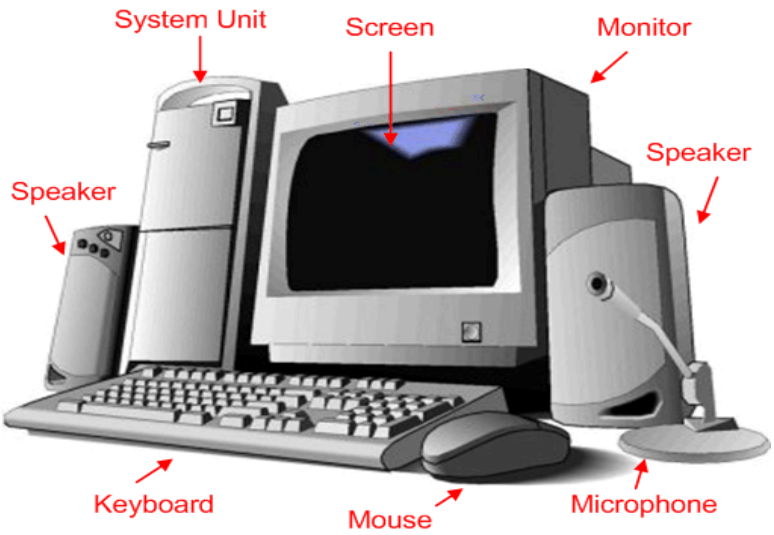
Consists of:

1-Hardware 2-Software 3-Users

Hardware

-Physical tangible parts of a PC ie can see or feel

-System unit & parts inside, monitor, keyboard, mouse etc.



Software

-Intangible i.e. cannot see or feel.

-Programs (instructions for controlling a PC).

Users

Consists of:

1-human ware (people) 2-machines

Human ware includes:

- ✓ End users, Programmers, System Administrators, Network Administrators, Database Administrators etc

Machines include*:*

- ✓ Printers, scanners, fax machines, digital camera etc.

Classification of hardware

1. *CPU*

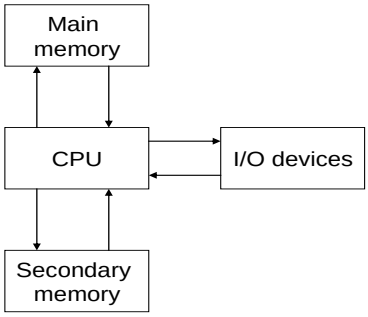
   Central Processing Unit

Considered brain of the computer
The main microchip in the computer

2. *Main memory*
   Primary storage space for programs & data
   Mainly RAM (Random Access Memory)

3. *Peripherals*
   Any hardware other than CPU & main memory
   Secondary memory & I/O devices

Simplified diagram



Note:
-Main hardware components are CPU & main memory
-Hardware can also be classified as
1. CPU 2. Memory 3. I/O devices

Memory
-The storage space for programs & data.
-Consists of cells with addresses.
-Of 2 main types:
1. Main memory 2. Secondary memory

Cell
-The smallest size of memory; of capacity 1 byte.
-Stores one character i.e. alphanumeric, symbol etc.

Cell address
-The location of a cell in memory.
-CPU uses it to get information for its functions.

1.1 Introduction to I/O devices

Input (I/P) devices
-Used to enter data into the computer
-For example: keyboard, mouse, scanner, microphone, digital camera

Output (O/P) devices
-Used to get information out of the computer
-For example: monitor, speaker / headphone

Ports & connectors
-Also known as **peripheral device interfaces.**
-The various connectors and ports allow the PC to communicate with the many different devices and peripherals attached to it.
-Used to connect I/O devices to system unit and to transfer data to and from them.
-The two fit together to connect I/O device to PC
-Most are located at the rear of the system unit
-Classified as: 1- female or male 2- serial or parallel
-The types are:
PS2 (mini DIN-6), DB, BNC, RJ, USB, FireWire (IEEE 1394) etc.

Serial port & **connector**

- Designated as COM1, COM2 etc.
- Serial communication is the simplest form of data transfer in modern PC ie transmits one bit (0 or 1) along a wire (line) at a time.

- In modern PCs, serial communication is **full duplex** i.e. both of the communicating devices can send data at the same time.

Compare full duplex with half-duplex.

Advantages

1. Less number of wires; can use longer cable.
2. Facilitate communication of PC with external device.

Parallel port & connector

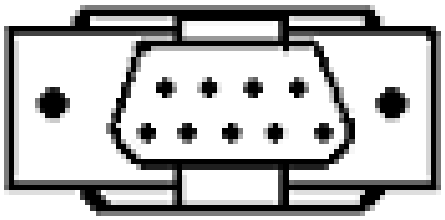
- Designated as LPT1, LPT2 etc.
- -Transmit more than one bit at once i.e. several lines each with I bit to transmit.
- -A standard parallel connection on a PC transmits 8 bits (1 byte) at a time, with the total bandwidth of 12 Mbps.
- -Originally designed for use by printers, the parallel port has seen many devices that take advantage of this interface on PCs.
- -Common uses on the parallel port range from printers, external hard drives and CD-ROMs.
- -A parallel port on a PC uses a **DB-25 female connector**. When connecting a printer through a parallel port, a special cable with a **DB-25 male connector** on one end is used.

Disadvantages

1. Attenuation i.e. signal becomes weaker.
2. Thermal noise & cross talk effect.

DB port & connector

DB-25 female

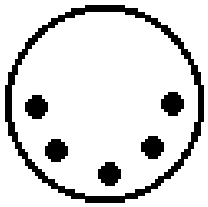


DB-9 female

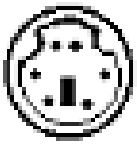
-DB stands for **Data Bus**
-They are D-shaped & may be female or male
-Female has holes & male has pins
-No. of holes or pins are commonly 9, 15 or 25
i.e. we have DB-9, DB-15 & DB-25.

-May be serial or parallel ports e.g. DB-25 is parallel.
- DB male connector port at the back of PC (DB-9 male or DB-25 male) is a serial port.
-Serial ports are also called **COM ports** or **RS-232** ports (Reference Standard #232 by IEEE).
-DB-25 female connector at the back of PC is a parallel port, commonly used to connect printers, scanners & backup systems (e.g. external hard drives).
-DB-25 female connector, originally also connected keyboard & mouse.
-DB-25 female of 3 rows (each of 5 holes) is for VGA or SVGA video monitor adapter, but DB-25 of 2 rows (of 8 and 7 holes), may be for playing games (joystick adapter).
See DB-25 female diagram above with 3 rows (each of 5 holes).

PS/2 port & connector



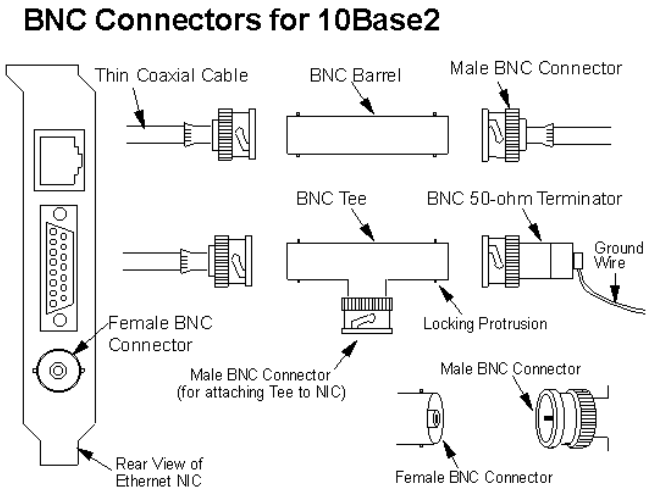
AT (DIN-5)



ATX (DIN-6) developed in 1987 by IBM for keyboard and mouse of IBM

-PS/2 stands for **Personal System for IBM Series 2** model.
- Introduced by IBM for keyboard and mouse of IBM compatible PCs, to replace DB-9 for older PCs.
-It not hot-pluggable i.e. first switch off the PC to install
-Of 2 technologies:
1. obsolete AT (Advanced Technology) or DIN-5 (5 pin)
2. Still used ATX (Extended Advanced Technology) or DIN-6 (6 pin).
- AT (DIN-5) introduced in 1984 and ATX (DIN-6) in 1987 by IBM.

BNC port & connector

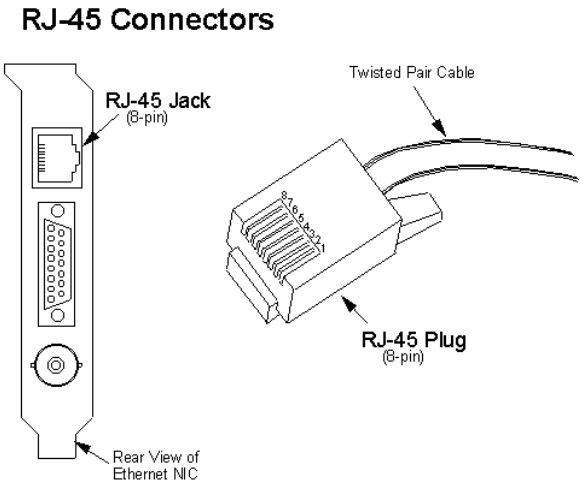


-BNC stands for **British Naval Connector.**

-Circular in shape, like Nokia mobile charger port.

-Found on sound card, to connect e.g. speakers, earphones & microphone

RJ port & connector



-RJ stands for **Registered Jack**, used for communication devices.

-Of 2 types RJ-11 (smaller) for phone line and RJ-45 (bigger) for modem (*Mod*ulate-*Dem*odulate) or NIC (Network Interface Card) connection.

-Numbers 11 and 45 do not indicate number of pins or holes!

-RJ-11 connects 4 or 6 wires, but RJ-45 connects 8.

USB port & connector

-USB stands for **Universal Serial Bus**, the standard port for many new PCs.
-Used to connect a wide range of devices to PC i.e. flash, keyboard, mouse, printer, scanner etc.
-It is rapidly replacing serial and parallel communications.
-USB was developed by a group of industry giants such as Compaq, IBM, Intel, Microsoft, NEC and Nortel Networks.
-Originally intended as an interface for low bandwidth devices e.g. modems, keyboards, mice and printers
-Many hardware manufacturers now use the USB interface for other devices such as CD-RWs and external hard drives.
-It is **plug and play** i.e. it allows you to connect device without configuration.
-It also allows **hot swapping** or **hot plugging i.e.** it allows you to connect device while PC is running.
-The old USB 1.1 specification allows for **127 devices** to be attached to one USB interface, with an overall bandwidth of 12 Mbps that is shared by all the devices in the chain.
-The new USB 2.0 (HS-USB) specification increases the bandwidth from 12 Mbps to 480 Mbps, and is backward compatible with USB 1.1 devices.

Advantages of USB

1. Faster communication
2. Supports plug and play and hot swapping
3. USB hub can provide power to few low power peripherals

See hard drives later.

FireWire (IEEE 1394)

-Originally developed by Apple Computer under the trademarked name is for high bandwidth peripheral devices.
-IEEE 1394 supports **63 devices** in a chain, with up to 400 Mbps of shared bandwidth for the devices.
-IEEE 1394 (like USB), also supports **hot swapping** of devices.
-Current uses of IEEE 1394 are commonly seen in the digital video cameras, where high bandwidth data transfers are necessary.
-Many developers are using this technology for other devices eg external hard drives, CD-RWs, and other bandwidth intensive devices.. -Apple computers and some PC based laptops come with integrated IEEE 1394 technology.

Common I/O devices

These include: Keyboard, Mouse, Monitor, Printer and Scanner.

Keyboard

-Primary I/P device & control device of a PC
-Data & commands are entered through KB by typing
-Basically consists of: type writer-like keys, function keys & numeric keypad keys
-Of 2 styles: obsolete 5 pin AT & still used 6 pin ATX.

Connection of Keyboard to PC

1-DB-9 (obsolete) 2- PS/2 (older) 3- USB (newer)

Pointing devices

-These are I/P devices, commonly used to enter commands to PC
-They include: mouse, joystick, light pen etc
-Others found on laptops, equivalents of mouse are: touch pad, track ball & track stick.

Mouse

-Main device used to give commands to PC
-Has basically left & right buttons
-Left button used to select, choose (click), double click & drag items on the screen
-Right button used to click & select options
-*Intel mouse* has a roller for scrolling up & down the document.

Connection of mouse to PC

1. DB-9 (obsolete) 2. PS/2 (older) 3. USB (newer)

Wireless mouse

-This is the latest technology of mouse. Also called **cordless mouse**.
-Of 2 types:
1. Infrared (I.R) 2. Radio frequency (R.F)
-R.F moves around corners (as sound waves) but I.R travels in a straight line
-Therefore R.F is better than I.R, since no need to align mouse to PC.

Detection types of mouse

1. Mechanical 2. Optical 3. Laser

Mechanical type

-Oldest type still in use
-Has a rubber ball at bottom, which needs a pad to roll on

-Cheapest & most common type of mouse (by then)
-Needs cleaning due to dust & other particles

Optical type
-Newer type with optical sensor technology.
-Uses reflected red LED light to detect action.

Laser type
-Latest type of mouse on the market.
-Uses laser beam of light to detect action.

Monitor

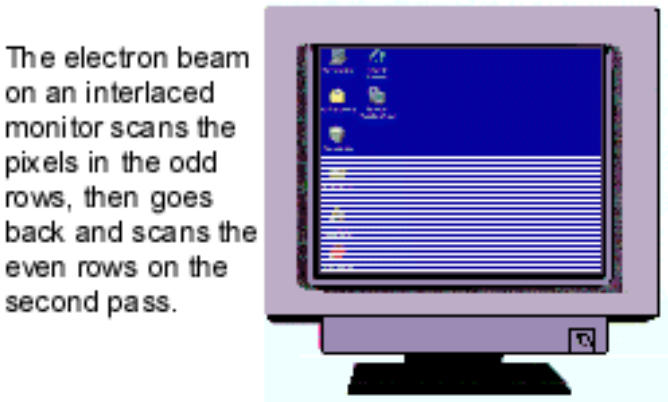
-It is mainly output device that is part of your computer's display system but can also be used as input device e.g. when using light pens to draw.
-It is used to see what is happening inside the PC
-**Soft copy** is what is seen on screen or stored on secondary storage media.
-It has the screen outside and other components inside.
-A cable connects the monitor to a **video adapter (video card)** that is installed in an **expansion slot** on the **motherboard**.
-Of 3 main technologies:

1. CRT (older) 2. LCD (modern) 3.TFT (latest)

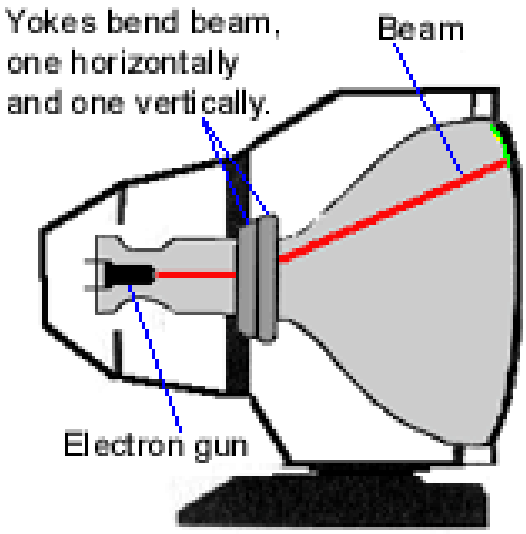
Note:

- CRT (Cathode Ray Tube) is bulky & heavy
- LCD (Liquid Crystal Display) has a flat screen & it is light
- TFT (Thin Film Transistor) is modern special type of LCD.

Monitor screen



CRT (Cathode Ray Tube)



Display types (technologies) of monitor

1. Monochrome (obsolete)- one color & 2 dimensional
2. Colored display (obsolete)- many colors & 2 dimensional
3. Graphic displays (modern)-3 dimensional

-Graphic displays include:
EGA (Enhanced Graphics Array) & VGA (Video Graphics Array)
-These are incorporated in the technology of the *video cards (discussed later).*

Pixel (picture element)
-Means the smallest spot on the display (screen).

-A graphic (diagram or picture) on a monitor screen consist of thousands of tiny dots called **pixels**.

Connection of monitor to PC
-Uses DB-15 to connect to PC
-Female on PC & male on monitor

What to look for in monitors
1. Screen size –measured diagonally in inches e.g. 15'' or 17''
2. Resolution –expressed as two numbers.
3. Display types (standards)- graphic displays i.e. monochrome, CGA/EGA, VGA, SVGA, XGA.

Resolution
-It is the number of pixels that are used to draw an image on the screen.
-Expressed as two numbers i.e. horizontal (row) pixels and vertical (column) pixels e.g. resolution of 800 X 600 = 480,000 pixels.
- A monitor must be matched with the video card in the system, the right resolution and colors are displayed.
-For example, if the monitor is capable of displaying a resolution of 1,024 X 768 but your adapter has resolution 640 X 480, then you get the later.

Note:
-The different technologies refer to a combination of the video card and monitor.
-Display types (standards) describe the basic number of colors and resolutions for each technology.
Monochrome (obsolete)
-Monochrome monitors are very basic displays that produce only one color.

CGA/EGA (disappearing)

-Color Graphics Adapter (CGA) introduced color to the personal computer. In APA mode it can produce a resolution of 320 X 200 and has a palette of 16 colors but can only display 4 at a time.
-Enhanced Graphics Adapter (EGA) by IBM, the proper monitor was capable of a resolution of 640 X 350 pixels and could display 16 colors from a palette of 64.
-There were three electron guns, one for each primary color ie red, green and blue (RGB). This combination could produce 8 colors.

VGA (still in use)

-Video Graphics Array (VGA) by IBM, capable of a resolution of 640 X 480 pixels and could display up to 256 colors.

-This technology soon became the standard for almost every video card and monitor being developed.

SVGA (modern)

-With Super Video Graphics Array (SVGA), manufacturers began to develop video adapters that added features and enhancements to the VGA standard.

-Super-VGA is based on VGA standards and describes display systems with several different resolutions and a varied number of colors.

-At first SVGA could had capabilities of 800 X 600 with 256 colors or 1024 X 768 with 16 colors. However, these cards and monitors are now capable of resolutions up to 1280 X 1024 with a palette of more than 16 million colors.

XGA (modern)

-Extended Graphics Array (XGA) was developed by IBM. It improved upon the VGA standard (also developed by IBM) but was a **proprietary** (not standard) adapter for use in Micro Channel Architecture expansion slots.

-XGA allowed for better performance, could provide higher resolution and more colors than the VGA and SVGA cards at the time.

-However, it was only available for IBM machines, and many of its features were later incorporated by other video card manufacturers.

Printer

- It is output device used to produce hard copy i.e. what is printed on paper (printing, envelop, sticker, business card etc), envelope, transparency sheet (car number plates), T-shirt etc

Connection of printer to PC

-Uses DB-25 to connect to PC, female on PC & male on printer.

-Modern printers can even use USB ports.

What to look for in a printer

1. Printing speed- measured in cps (character per second)
2. Resolution- measured in dpi (dots per inch)

The 2 main types of printers:

1- Non-impact printers 2- Impact printers

Non-impact printers

- Use chemical, ink, toner or heat (thermal) i.e. inkjet (desk jet), laser jet & thermal printer

Impact printers

- Use small hammers (pins) to print i.e. dot matrix, daisy wheel.

Printers can also be classified as;

1-serial or parallel printers 2-character, page or line printers

Some printers are specialized

-For example, *plotters* convert graphics, charts & drawings into precision hard copy.

-Plotters used by engineering, architects, physicians etc.

Brands of printers

Canon, HP, Epson, Xerox etc.

Showing 2 forms of printers

Scanner

-Also known as **image scanner** is an input device, taken to be the "eye" of the PC.

-It is a device that optically scans images, printed text, handwriting or an object, and converts it to a digital image.

-Can see the images (picture, photographs etc) or printed text sent into PC memory

-Image can be pasted in text or edited & printed e.g. magazines, newspapers, textbooks

-Some are special scanners e.g. **OCR** (Optical Character Recognition).

What to look for in a scanner

1. Scanner resolution- measured in dpi (dots per inch)
2. Optical resolution- about color each dot holds

Types of scanners

1. Flatbed (desktop)
   Document placed on a glass window for scanning
2. Sheet fed (document scanner)
   Paper is pulled through scanner like fax machine.
3. Hand held
   The device is moved by hand over the object scanned.

Note:

1. Depends on what is to be scanned eg books use flatbed, large stacks of pages use sheet fed, labels on items use hand held.
2. You can have three in one ie printer, scanner and photocopier.

Flatbed scanner with cover closed (left) and cover open (right)

MODEM (Modulate-Demodulate)

-Enables a PC to communicate via telephone line, for Internet connections.

- Can be internal or external or wireless.

Types of modems

1. Dial up Modem inside older computers.
2. Cable Modem for coaxial cable as that of cable TV (not in Uganda)
3. DSL (Dial Subscriber Line) Modem for telephone lines.
4. Broadband Modem

Internal modem

External modems

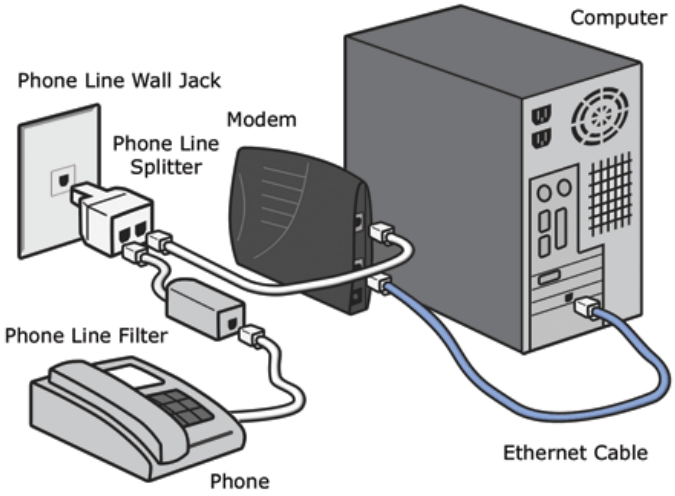
Internal modem

- It is expansion card (circuit board) fitted in expansion slots on the motherboard inside the system unit.
- It allows the computer to communicate over a network with other connected computers.
- Has two RJ-45 ports:
  1. PHONE connects to telephone 2. LINE connects to PC
- Of 2 types:
  1. dial-up 2. Wi-Fi (Wireless-Fidelity)
- Dial-up operates over a telephone line and requires a network access. Phone number and logon credentials to make a connection.
- Wi-Fi connect wirelessly and without credentials in certain cases.

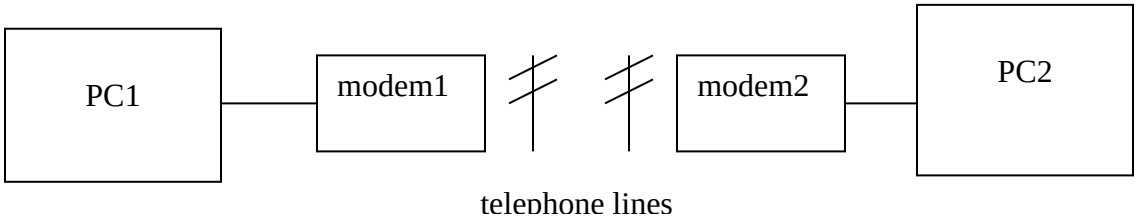
External modem

- It is enclosed in separate case and not inside the system unit.
- The modem is connected to the computer and telephone line by means of two separate cables.

How to connect external modem



How a modem works



-The sending modem 1 modulates (converts the digital signal from the PC1 into analog signal for the telephone system).
-The reason for this is that the telephone system was originally designed for voice signals (analog signals).
-The receiving modem2 demodulates (converts the analog signal from the telephone system back to digital signal for the PC2).

1.2 Introduction to CPU

- CPU stands for **Central Processing Unit**.
- The main microchip (simply chip) or **IC** (Integrated Circuit).
- Found inside system unit installed on the motherboard.
- Considered the brain or heart of the PC.
- CPU also known as microprocessor or simply processor, for only PCs.
- CPU is just part of the microprocessor, for other digital systems
  i.e. **PLC** (Programmable Logic Controllers)
  PIC (Programmable Interface Controllers).

Manufacturers of common (popular) CPUs

-The 2 common (popular) brands are:

1. **Intel** for mainly IBM compatible PCs
2. **AMD** (Advanced Micro Devices)

- Intel and AMD make most of CPUs for Window-based PCs.
- Other manufacturers (not common or popular) exist e.g. Cyrix, Motorola.
- CPUs are manufactured from semiconductors e.g. silicon and Pentium.
- The 2 main types of interface (connection) to motherboard are:

1. Socket 2. Slot

Note:

- Sockets dominated for many years (up to socket 8 in 1997).
- Manufacturers then changed to slot (1998-1999) i.e. short-lived.
- Manufacturers then changed back to socket after 1 year only.

- Today it is a mixture of both socket and slot, however socket is more common (popular).
- **Intel** CPUs include Standard Pentium (1993), Pentium Pro (1995), Pentium II (1997), Pentium II Xeon (1998), Pentium III (1998), Pentium III Xeon (1999), Pentium MMX(1998), Celeron (1998), Pentium IV (2000), Italium.
- Pentium MMX (Multimedia Extensions) was first Intel CPU designed to handle multimedia (audio and video).
- Celeron was designed to be a low-cost alternative to Pentiums, because of lower **cache memory** and lower speed than Pentiums.
- First Celeron was based on the Pentium II, but subsequent Celerons were based on the Pentium III, Pentium IV, Pentium MMX, and Core 2 Duo branded processors.

See modern Intel processors (November, 2014) ie Cores i3, i5 and i7.

- **AMD** CPUs include Athlon, Thunderbird, Duron series and Spitfire.

Socket interface

- Notable (now obsolete) socket types are:
  - Socket 1 (169 pins, 5 V, 133 to 200MHz)
  - socket 5 (320 pins, 3.3 V, 75 to 133 MHz)
  - Socket 7 (321 pins) for Pentium MMX, AMD K5 & K6.
  - Socket 370 (370 pins) for Pentium III & some Celerons.
- Socket 370 replaced Slot 1 CPU interface on personal computers.

Left: Intel Socket 7 Right: Interface of Intel Socket 7 on motherboard

Slot interface

- **SECC** (Single Edge Contact Card) and **SEPP** (Single Edge Processor Package) refer to slot type.
- They have pins on only one side & are mounted perpendicularly into motherboard slot.
- Notable (now obsolete) slot types are:
  - Slot 1 (1997) SECC by Intel Company, supported Pentium II & III.
  - Slot A by AMD was short-lived i.e. AMD Athlon.

Design considerations of CPU

- Refers to motherboard-CPU Compatibility.

1. Power consumption (voltage) – As low as possible (initially 5 V).
2. Interface to motherboard –Easy insertion i.e. ZIF for socket types.
3. CPU speed –In MHz (obsolete CPUs) or GHz (modern CPUs).
4. Heat reduction –Provide electric fan & heat sink for cooling CPU.

Cooling fan & heat sink on top of CPU

CPU Packaging

There are different form factors (sizes & shapes) of CPUs.
The 2 most commonly used today are:

1. **PGA** (Pin Grid Array)
2. **SECC** (Single Edge Contact Cartridge).

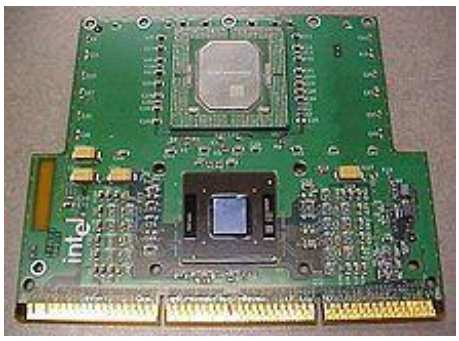
PGA (Pin Grid Array)

PGA (socket) CPUs: Left (Intel Pentium) & Right (AMD Athlon)

- PGA packaging is the most common (popular) and can be easily identified.
- They are flat square microchip packaged in ceramic or plastic.
- They are known as **PGA** (Pin Grid Array), because of array of pins underneath.
- They differ in the number of pins, power consumption (voltage) and speed (MHz for obsolete or GHz for newer type).
- Most PGA processors plug directly into the motherboard using **ZIF** (Zero Insertion Force).
- ZIF sockets typically have a metal arm on one side used to clamp the processor down in place.
- PGA processors have to be carefully aligned to fit into the ZIF sockets to prevent bent pins, which could damage the processor.

Left: Lever up on CPU socket Right: Align 2 similar corners (without pins)

SECC (Single Edge Contact Cartridge)



SECC (slot) CPUs: Left (Pentium II) & Right (Pentium II Xeon) without cover

- The **cartridge processor** is put in a SECC package and concealed within a protective plastic cover.

- The outside has markings (labels) on the case and the contacts that connect into the slot on the motherboard.

<See diagrams of SECC (slot) CPUs>.

- A similar microprocessor is **SEPP** (Single Edge Processor Package).
- **SEPP package does not have a protective plastic cover as does the SECC.**
- SEPP is associated with Intel Celeron processors and of the two, SECC is now more common.
- SECC was introduced with Pentium II range of processors.
- SECC processor is plugged into a slot on the motherboard just like an **expansion card**.

Intel Pentium II SECC (Slot 1) plugged on motherboard

CPU has 3 main components:

1. Control Unit
2. ALU (Arithmetic Logic Unit)
3. Registers

Control Unit

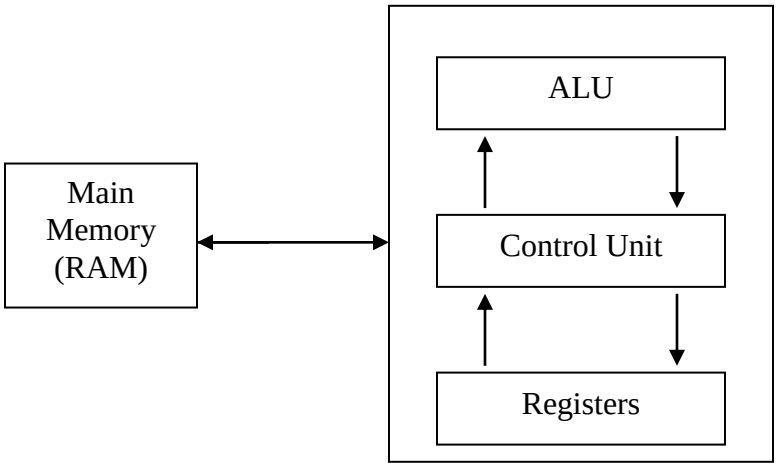
- Coordinates processing i.e. fetch-decode-execute.
- Regulates the flow of information through the processor.

ALU (Arithmetic Logic Unit)

- Handles logical & arithmetic operations i.e. decisions & calculations.
- Executes instructions to various peripherals e.g. I/O devices.
- Also used for maintaining timers that help functioning of PC.
- CPU can have more than one ALU.

Registers

- Very small memory locations (microchips) in CPU for read & write.
- Responsible for holding the data that is to be processed.
- Used to store temporary results & control information.
- High speed memory and so most expensive memory in computer.
- Has instruction pointer (directs the CPU to the next memory location) & accumulator (for storing the next values).

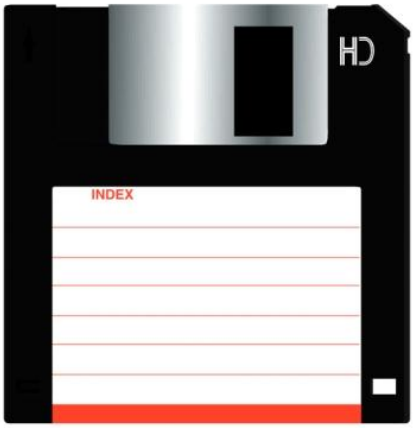


1.3 Computer storage devices

- They refer to computer components which can retain the data in digital format, to be used for computing at some point of time in the future.
- They have memory (storage space), and can be classified as:
    1. Permanent (long term) or temporary (short term)
    2. External (portable) or internal (not portable)
    3. Magnetic (stores magnetically) or optical (stores optically).
- Optical disks use laser beams and are of 2 common types:
    1. CD-ROM or simply CD (Compact Disk)
    2. DVD (Digital Versatile Disk)
- Magnetic disks are of 2 common types:
    1. Floppy disk
    2. Hard disk drive.

-Disk storage refers to floppy disks, hard disks and optical disks, where data is recorded on a round and rotating surface. Disk drives are used to record or retrieve information from these disks.

Floppy Disk



- Invented by **IBM in 1969**, also known as diskette or just floppy.
- **It is external, permanent and magnetic storage device.**
- A floppy disk is read and written by a floppy disk drive (FDD).
- Has had 3 sizes 8-inch, 5¼-inch and 3½-inch (most common).
- Has storage capacity of 1.5MB or 1.44MB.
- Used to be main medium of copying from PC e.g. for backups.
- Floppy disks enjoyed wide popularity from the 70s to the 90s.
- Has been replaced by devices which had more capacity and faster e.g. flash disk i.e. has become almost absolute.

Hard Disk Drive

Internal IDE hard disk drive showing the label and interfaces

- Invented by an **IBM team**, led by Rey Johnson on 14th December, **1954**.
- Also known as just **hard disk** or **drive C (C drive)**.
- **It is internal or external, permanent and magnetic storage device.**
- It the main storage device with the biggest capacity.
-First hard disk had a capacity only 5MB, but today the capacity has increased to hundreds of GB.
- Used to store digital data on rotating platters having magnetic surfaces.
- Stores OS (Operating System), applications (packages) and user's files e.g. word documents, spreadsheets, databases, music.

Note:
Slow PC may mean HDD is full (more than 90% full).

Compact Disk (CD)

- Compact disk, more commonly referred to as a **CD**, is an **external**, **permanent**, **optical disk** used to store digital data.
- Introduced in 1982, and later followed by CD-ROM, CD-R, CD-RW, VCD, SVCD and Enhanced CD.
- CD-ROM and CD-R are widely used in the field of computers.
- CD-R (Ready only) or CD-RW (Rewritable /Read & Write) are read / written by CD-ROM drives and DVD drives (internal or external).
- Originally, CD-ROMS were read only devices, but now read/write technology has been developed.
- Can hold OS for booting if hard disk has a problem, but it is slower than hard disk.
- Also used to hold other SW e.g. applications (packages).
- A standard CD having a diameter of 120mm can hold up to 700MB data.
- Drive speed is the number of times the drive is faster than the standard CD-ROM reader i.e. speed of audio CD player = 600 kbps.
  e.g. speed 52x = 52x600KBps.

- Read/Write speeds are expressed as **read**, **write** or **rewrite**.

See interface to motherboard later Chapter 5 –Drives.

Digital Versatile Disk (DVD)

DVD-R (Read only) disc

- Digital versatile disk, more commonly referred to as **DVD**, is an **external**, **optical storage** device used for data storage in computer technology.
- Although they share the same dimensions as the CDs, they have the capacity to store much more data.
- DVD-video is used to store video (pictures), while DVD-audio is used to store audio (sound).
- DVD-R (Ready only) or DVD-RW (Rewritable /Read & Write) are read / written by only DVD drives (internal or external).
- Originally, DVDs were read only devices, but now read/write technology has been developed.
- Just like CDs, they can hold OS for booting if hard disk has a problem, but it is slower than hard disk.
- Also used to hold other SW e.g. applications (packages).

Zip drive

Zip disk media

Internal Zip drive

- A removable cartridge storage device that may be used to store compressed data as a data backup method.
- The Zip drive is a medium-capacity removable disk storage system that was introduced by **Iomega in late 1994**.
- Originally, Zip disks launched with capacities of 100 MB, but later versions increased this to first 250 MB, then 750 MB and now up to 2GB.

Tape drive

- It is one of the most efficient and easiest ways for backup.
- The primary use of a tape drive is to store important information present in a computer.
- The tape drive uses a magnetic tape and can be external or internal.
- It can have the largest capacity compared to all other storage devices.
- They are very slow (i.e. backup can take a long time) and the tapes are expensive.

Jaz drive

Internal and external 1GB Iomega Jaz drives with media

- The Jaz drive was a removable disk storage system, introduced by the **Iomega Company in 1995**. i.e. now obsolete.
- The Jaz disks were originally released with a 1 GB capacity in a 3½-inch form factor, which was a significant increase over Iomega's most popular product at the time, the Zip drive with its 100 MB capacity.

USB flash drive

- A USB flash drive consists of a flash memory data storage device integrated with a USB (Universal Serial Bus) interface.
- USB flash drives are typically removable and rewritable, and physically much smaller than a floppy disk.
- Some allow 1 million write or erase cycles and offer a 10-year shelf storage time.
- USB flash drives are often used for the same purposes for which floppy disks or CD-ROMs were used.
- They are smaller, faster, have thousands of times more capacity, and are more durable and reliable because of their lack of moving parts.

Memory Card

- A memory card or flash card is an electronic flash memory data storage device used for storing digital information.

- They are commonly used in many electronic devices, including digital cameras, mobile phones, laptop computers, MP3 players and video game consoles.
- They are small, rewritable and they can retain data without power.

See memory cards below.

Secure Digital card (SD)

MiniSD Card

CompactFlash (CF-I)

Memory Stick

MultiMediaCard (MMC)

SmartMedia

xD-Picture Card (xD)

- These are the internal storage spaces (memory) of a PC found inside the system unit.
- There are 2 main types:
    1. RAM chips
    2. ROM chips

RAM (Random Access Memory)

- It is the working space of the PC i.e. data and running programs (processes) are stored here temporarily.
- Random Access means any point of this memory can be reached directly, as compared to sequential access for tape storage device.
- Data and SW are stored on drives, normally the hard drive. However, it is easier for to read from and write into RAM.
- Characteristics of RAM:
    - Temporary memory (volatile) i.e. lose information when power is off.
    - Possible to read from and write into RAM i.e. change or add data.
    - RAM chips can be upgraded i.e. changed (removed and added).

<See different types of RAM & RAM modules later e.g. CMOS RAM>.

ROM (Read Only Memory)

- Read only means the computer only reads from ROM i.e. impossible to change or add data.
- Characteristics of ROM:
  - Permanent memory (non-volatile) i.e. retains information even when power is off.
  - Impossible to write into ROM i.e. cannot change or add data.
  - ROM chips are not upgraded, since they are soldered on motherboard.

1.4 The System board

- Commonly known as the motherboard, is the main PCB (Printed Circuit Board) found inside the system unit.
- Referred to as the "*nervous system"* of the PC, since it enables the devices to communicate with the CPU.
- It holds and connects the PC's electronic pieces e.g. CPU, RAM modules, ROM chips, buses, controllers, connectors & ports, expansion cards, expansion slots etc.

1.5 Expansion cards

- These are smaller PCBs that fit into the MB in expansion slots.
- They add functions to the PC i.e. give added capabilities.
- Common expansion cards are:
  1. Sound card -To add sound function e.g. speakers, microphone.
  2. Video card -To handle video, graphics, pictures etc.
  3. NIC -To enable a PC connect to a network.
  4. MODEM -To enable a PC communicate via Internet.

NIC (Network Interface Card) & MODEM (Modulate-Demodulate).

Sound cards

-They perform several tasks on sound i.e. recording, saving on a hard disk, mixing, editing and playing sound.

-They can also support audio compression; also used by HDTV (High Definition TV).

-Many are onboard cards (built in the motherboard), but can also be expansion cards (inserted into PCI slots in the motherboard). It is better to buy and install a modern sound card.

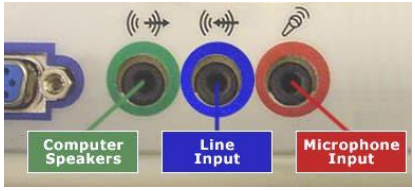
-They provide:

- Output ports for speakers & earphones
- Input ports for microphone
- Ports for CD & DVD players
- Other digital sound equipment

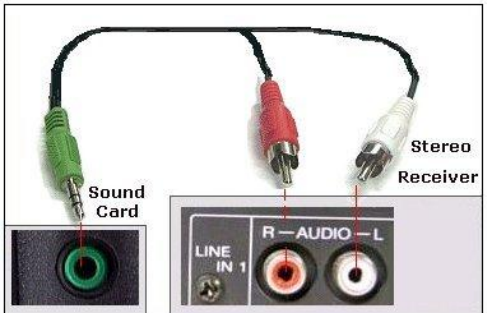
-For example, for music production use modern sound card for better sound quality and other features.

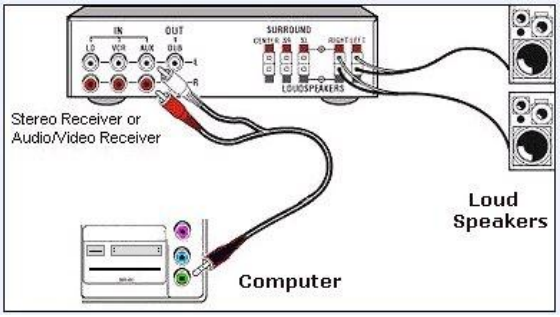
-Most sound cards have 3 BNC (round) color coded jacks as:

- Line out (**green** with arrow pointing outward) for speakers
- Line in (**blue** with arrow pointing inward) for CD player and cassette tape).
- MIC in (**pink** with picture of microphone) for microphone.



See connecting PC to a stereo receiver (radio) using Y adapter cables





Video capture cards & TV tuner cards

-Video capture cards decode (convert) analog video signal (e.g. from video camera or DVD player) to digital video for PC to process.

-TV tuner cards decode (convert) TV signals from TV channels (e.g. cable, aerial or satellite) for PC to process.

-These two are similar and so most:

- Video capture cards also function as TV tuners

- TV tuners function as video capture cards

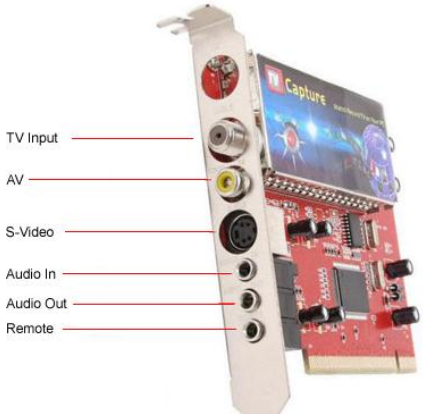
-These cards are installed into the PCI slots or the newer PCIe (PCI Express) slots in motherboard.

-Video capture devices without TV tuners also exist and are usually used in conjunction with external tuners.

Note:

Video capture cards are not video display cards (VGA cards) which provide output to the screen (monitor or TV).

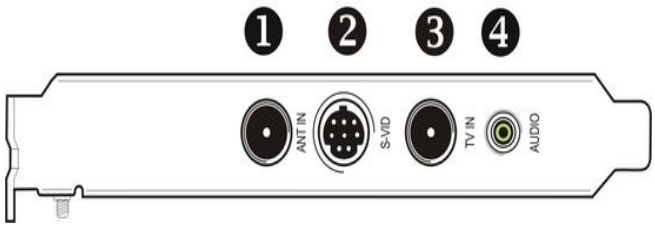
Video capture card



TV tuner card

Common ports are:

1. Antenna (coaxial connection)
2. Video connection
3. Cable/Antenna (coaxial connection)
4. Stereo audio mini jack connection



TV tuner ports

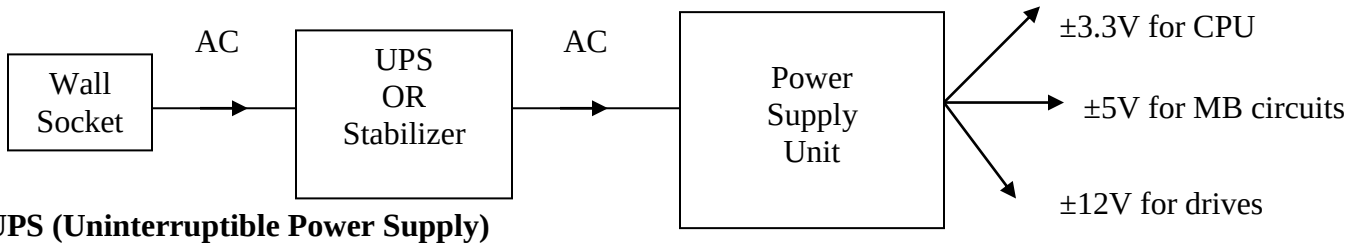
USB video adapter with audio

1.6 Electrical system

Consists of:

1. UPS or stabilizer
2. Power Supply Unit

3. Cables & connectors



UPS (Uninterruptible Power Supply)

-Has 2 purposes:

1. To regulate (stabilize) power in case of power fluctuations.
2. To provide backup power when power goes off.

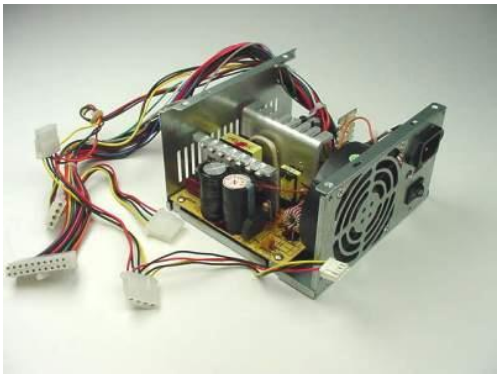
Stabilizer

- Only regulates (stabilizes) power, but does not backup.
- Has capacitors and diodes for smoothening.
- Some systems use backup battery which lasts more than UPS power.

Power Supply Unit

- Outside, it has power connectors (male and female), voltage selector (110/220V) & cables with Molex connectors (most common type).
- US standard uses 110V & 60Hz), but European standard uses 220V & 50HZ.
- Inside, it has step down transformers, rectifier circuit (diodes, resistors, capacitors & inductors), cooling system (fan & heat sink with metal fins) and fuses (to protect components).
- Standard 3-pronged power cord is used to connect Power Supply Unit to AC, for almost all PCs.

See diagram of Power Supply Unit next page.



Power supply: Outside Inside

Functions of Power Supply Unit are:

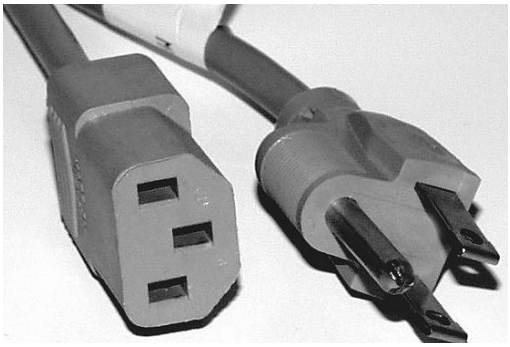
1. Steps down AC from 110/220V to various values required by the components.
2. Rectifies AC to DC (converts AC voltage from the wall outlet to DC voltage).
3. Distributes DC to various components according to their voltage requirements through voltage connectors.

Precaution!

- Set the voltage selector to correct voltage of the location (eg 220V for Uganda) before powering a computer.

Note:

- The 3.3V and 5V are used by digital circuits on MB (on board electronics), while 12V is used to run motors in disk drives (HDD, CD-ROM drive & FDD) and fans.
- Common power supplies come in a range of about 80 watts to 600 watts.
- Power is determined by the number of drives, the processor, number of expansion cards etc.



Standard AC power socket & plug

AC plug & wall socket

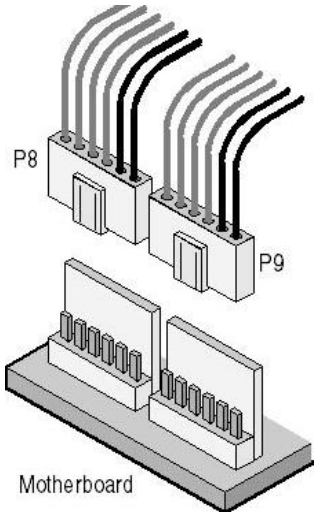
There are 2 main styles:

1. AT (Advanced Technology)
2. ATX (Extended Advanced Technology)

AT style

1. Molex connector to MB
   - For old PCs (microprocessors 486 & older Pentiums).
   - Has **single row, 2 connectors** for ±5V & ±12V.
   - Connectors labelled P8 and P9, each has a row of 6 pins.

Color code



Color	Voltage
Yellow	+12
Blue	-12
Red	+5
White	–5
Black	0 ground

2. Power Switch
   - Integrated directly into the power supply itself.
   - Turns the power supply on and off.
3. Wattage
   - Older ones provide a lower wattage rating than newer ones.
   - AT style usually provide wattage of less than 250W.

ATX style

1. Molex connector to MB

- For Pentium II and later PCs.
- Has **double row, single connector** for ±5V, ±12V & ±3.3V.
- Connector labelled P1 with 20 or 24 pins i.e. 10 or 12 pins per row.

20 pin P1 connector with colored cables

Color code

Color	Voltage	Pin	Pin	Voltage	Color
Orange	+3.3	1	13	+3.3	Orange
				+3.3 sense	Brown

Orange	+3.3	2	14	-12	Blue
Black	0 ground	3	15	0 ground	Black
Red	+5	4	16		Green
Black	0 ground	5	17	0 ground	Black
Red	+5	6	18	0 ground	Black
Black	0 ground	7	19	0 ground	Black
Grey	Power good	8	20	-5 optional	White
Purple	+5 standby	9	21	+5	Red
Yellow	+12	10	22	+5	Red
Yellow	+12	11	23	+5	Red
Orange	+3.3	12	24	0 ground	Black

The 20-pin omits the last 4 pins (shaded) ie pins 11, 12, 23 and 24.

3. Other connectors

20 pin Molex connector for MB

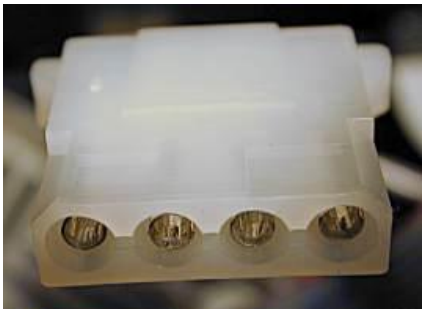
4 pin Molex connector for drives

4 pin Mini connectors

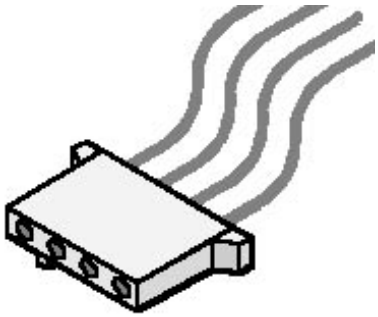
4 pin Molex connectors

- They are pin and socket types of connectors, named after **Molex Products Co**. that originally designed them.

- They were originally designed for use in appliances, but currently are standard in computer and automobile electronics.
- Male piece consists of 4 pins, while female piece consists of metal sockets.
- It provides both ±12V and ±5V e.g. for HDD, CD-ROM drives, DVD drives, internal tape drives.
- The Molex connector has two rounded corners and two sharp corners to ensure that it installs properly (see Figure below). It has four pins and cables which are larger than those of a mini connector.



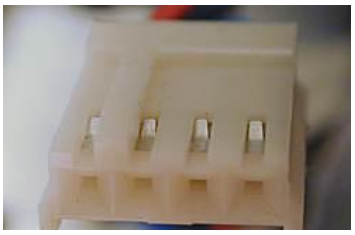
4 pin Molex connector



The 4 wires on connector

4 pin Mini connector

- Most power supplies provide one or more mini connectors.
- It is used primarily for 3.5-inch floppy disk drives (FDD).
- It has 4 pin-outs with usually 4 wires, and most have alignment key.
- Applying power with the connector reversed can damage or destroy the drive.



4 pin Mini connector

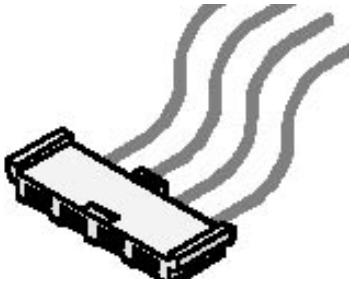


Diagram showing the 4 wires connected

2 or 3 pin Mini plugs

- A rare connector is used:
  - To connect the fan of Pentium II or III processor to MB for power
  - To connect CD-ROM drive to a sound card
  - To provide power for 3.5-inch floppy disk drives.

4. Power Switch

- Controlled by the motherboard and enables a computer supply to power off via software.

5. Wattage
   - Older power supplies provide a lower wattage rating than newer ones.
   - Newer ATX style power supplies normally provide 300W or more.

Extenders and splitters

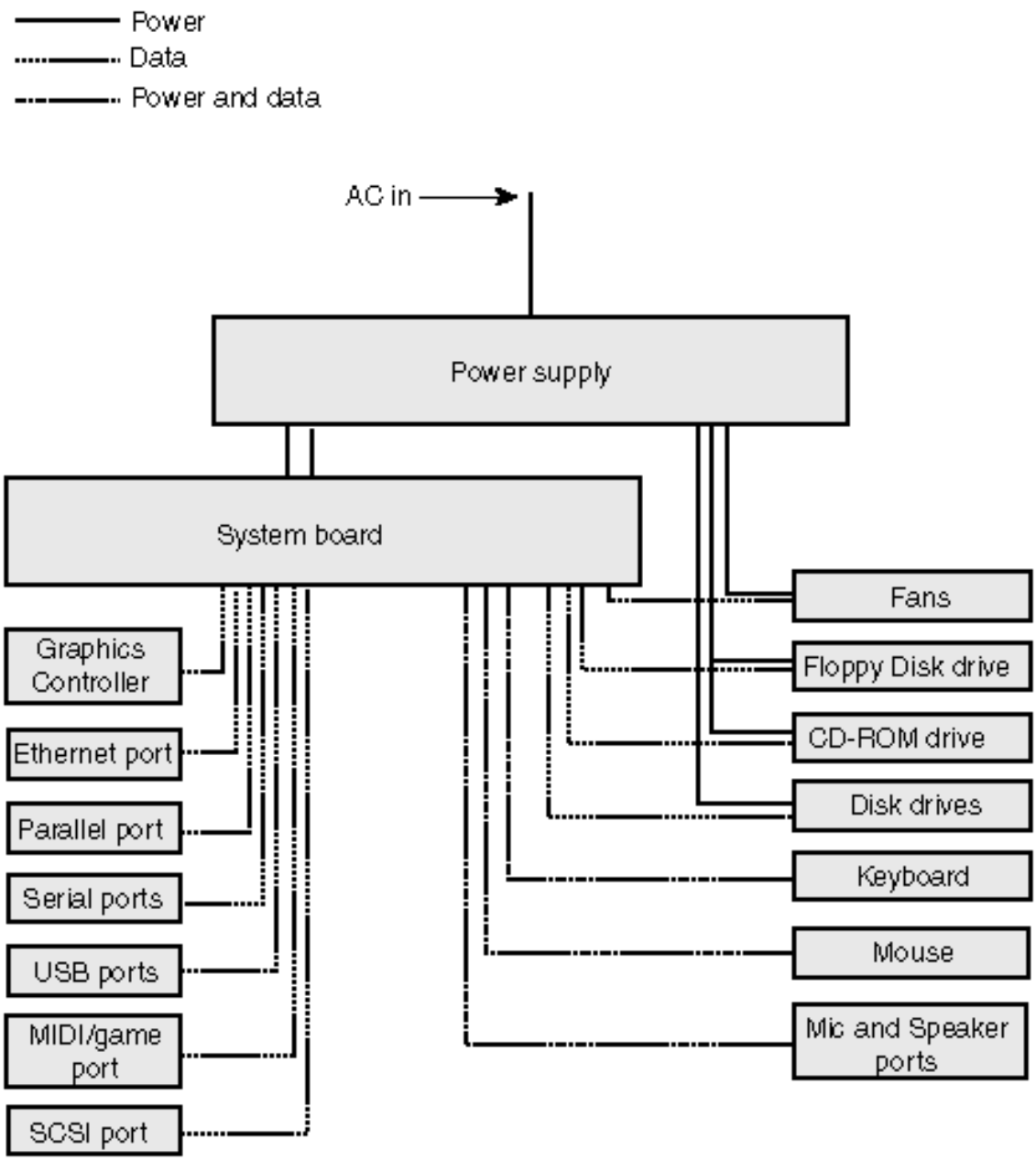
- **Extenders** are cables with Molex connector on each end, used to extend a power connection to a device beyond the reach of the power supply's own wiring.
- **Splitters** are similar to extenders, with the exception that they provide two power connections from a single power supply connector.

ATX Motherboard Power Extension Cable 4 pin Mini Splitter

What to consider in power supply

1. Form factor (size & shape) of power supply for the system unit.
2. Power rating (Watts) depending on how many devices/components.
3. Power connectors for the type of motherboard (AT or ATX).

Power and Data Signals



See troubleshooting Power Supplies later.

1.7 Computer Software

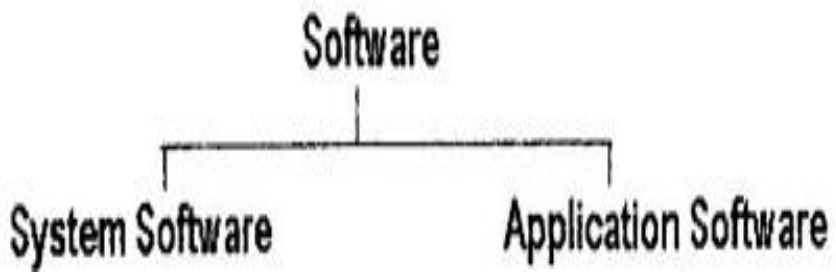
This Includes programs, documentation and configuration data.

Documentation is of 2 types:

1. System documentation -describes structure of the system.
2. User documentation -explains how to use the system.

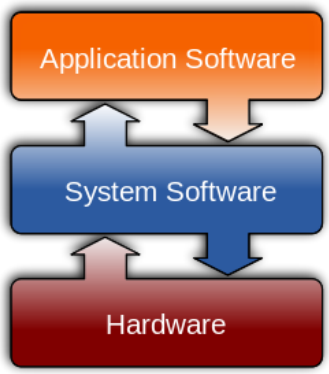
Software contains instructions for a computer to solve a specific problem.

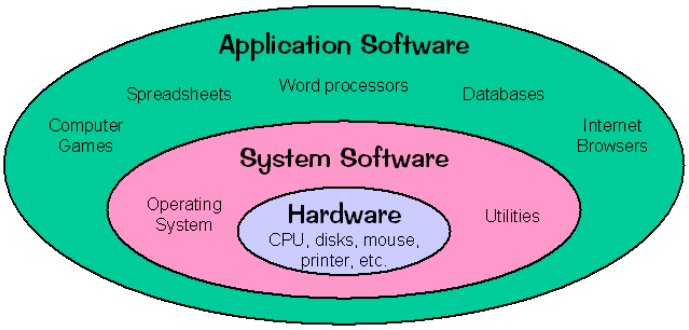
The 2 main types of software



System Software

- Software designed to control and coordinate the computer hardware and to provide a platform for running application software.





Note:

- Sometimes software can also be classified in 3 ways as:
  1. System Software 2. Application Software 3. Utilities
- System Software of 2 main types:
  1. Operating Systems 2. Utilities

OS (Operating System)

- This is the most important program that runs on a computer.
- It is the master program or manager of the computer system.
- It is the 1st program to be loaded into RAM when a computer is switched on.

Some of the functions of OS

1. Interface between user and hardware.
2. File and disk management.
3. Controls input/output devices.
4. Memory management (allocates memory to data & programs).
5. Task scheduling

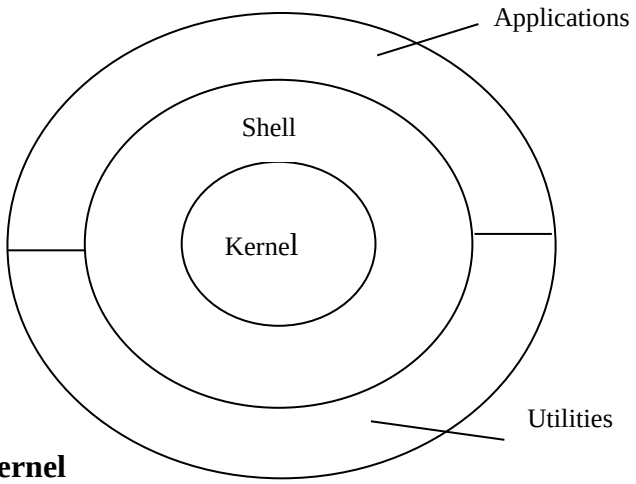
Some of the characteristics of OS

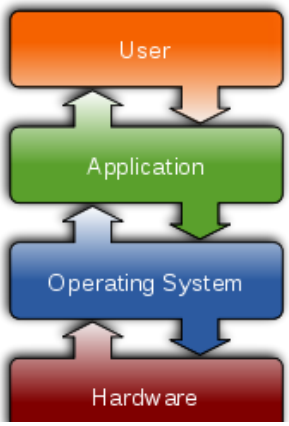
1. **Multi-user**
   Allows more than one user to run programs at the same time.
   Some allow hundreds or even thousands of concurrent users.
2. **User friendly**
   Easy to use e.g. has GUI (Graphical User Interface).
3. **Multi-tasking**
   Allows more than one program to run concurrently.
   Runs more than one program during a session.
4. **Multiprocessing**
   Supports more than one CPU e.g. dual core CPU.
5. **Multithreading**
   Allows different parts of a single program to run concurrently.
6. Real time operating system
   Responds to input instantly.

Note:
General-purpose operating systems (e.g. MS DOS, Windows and UNIX) are not real-time.
See Types of Operating Systems later.

The 2 main parts of OS

1. Shell 2. Kernel





Kernel

-This is the core of the OS i.e. coordinates all other parts of the OS.
-The master program of OS which enables the shell to work.
-1st to load into RAM and always present in RAM. Why?

-Makes it possible for user to interact with application SW.
-Uses LLL (Low Level Language) i.e. binary language.
-Responsible for loading & running programs (processes) and managing input & output.

Shell

- It is command interpreter providing command-line interface to users when entering commands, to access the services of the kernel.
- It makes it possible for the user to interact with applications (Packages & Utilities).

-The primary purpose of the shell is to invoke another program, but shells may have other tasks e.g. viewing the contents of directories.

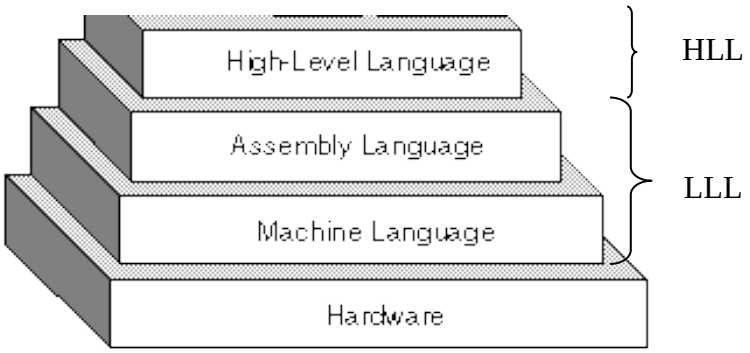
User interface
-The user interacts with OS through the user interface.
-This is part of OS used to issue commands by either typing at command line or using mouse at GUI (Graphic User Interface).

Utilities
-These are:
1. Utility programs 2.Programming languages 3. Programming tools

Utility programs
These are common purpose small programs used for limited capability e.g. drivers for devices, disk formatting, file management, disk defragmenting, disk cleanup & diagnosing problems.
Many come with OS (installed within OS), but others are installed separately from OS e.g. drivers are for both categories.

Programming languages
-Computer programmers use them to develop other software i.e. system software & application software.
-Think of them as bricks which are used to create other software (applications and operating systems).

-Programming languages are of 2 main types:
1. LLL (Low Level Languages) 2. HLL (High Level Languages)



LLL (Low Level Languages)
-They are machine dependent (written for one type of computer) e.g. LLL for IBM machines is not meant for Apple's Mac computers.

LLL are of 2 types:

1. Machine language
2. Assembly language

Machine languages
-Use binary bits (0s & 1s) and understood by computers.
-Every CPU has its own unique machine language.

Assembly languages
- Much easier to program, because they use **mnemonics** (string of letters).
- They are used to develop computer device drivers or other hardware interface development.

Note:

- Each different type of CPU has its own unique machine language.
- 4 GL (4th Generation Languages) are above HLL (closest to human languages).
- Program has to be converted to machine language for computer to understand.

HLL (High Level Languages)
-They are used by programmers to write commands (instructions) for computer hardware.
-Used to develop application software in software industries e.g. payroll calculation, ticket reservation, hotel reservation etc.
-Each HLL has a unique set of keywords (special words) and syntax (rules of language) for instructions.
-These include: BASIC, C, C++, C# (C sharp), Java, COBOL, FORTRAN, Ada, Pascal, Prolog, php & VB (Visual Basic).

-For example:

- C & C++ have been used extensively to develop operating systems.
- Java & VB (Visual Basic) are used to develop applications.
- PROLOG is used for artificial intelligence development.
- PHP, Java script & VB script are scripting languages used to develop Internet applications.

Programming tools

These are translators of LLL and HLL i.e. Assembler, Interpreter & Compilers e.g. **Compiler** changes **HLL** (source code) to **LLL** (object code).

Application Software

- Also known simply as **applications** or **software packages**.
- Developed to solve user problems or for a particular purpose.
- They are of 2 main types:
  1. Off-shelf 2. Tailor-made

Off-shelf

- Also known as **generic software**, are commercially developed to cater for a number of end users i.e. general in nature for all customers.
- Developed by SW company that also controls the SW specification e.g. Microsoft Corps.
- Many copies are sold on open markets by vendors. In a form to install & use immediately.
- Mainly software packages: e.g. word processors, spreadsheets, databases, accounting packages.

Tailor-made

- Also known as **bespoke** or **customized SW**, are made in-house (tailored to particular customers).
- Developed to meet specific needs of organization, customer or user
  i.e. developed according to customer's needs

Design issues of Software

-Refers to external & internal qualities of software.

1. Functionality i.e. produces required results.
2. Reliability i.e. always functional when required.
3. Usability i.e. Easy to learn, understand, operate etc.
4. Efficiency i.e. how fast it produces results.
5. Maintainability i.e. Changes on software can be done.

6. Portability i.e. can work under other HW & SW environments.

1.8 Types of Operating Systems

These include:

1. Windows Systems e.g. MS Windows
2. UNIX Systems e.g. Linux
3. DOS e.g. MS DOS

Note:

- CP/M (Control Program for Microcomputers) released in 1973 is no longer in use in modern computers.
- OS/2 (Operating System for IBM Series 2 models) released in 1992 was a collaborative project between Microsoft and IBM.
- Macintosh OS released in 2000 was used only with Apple computers, but not IBM compatible computers.

MS Windows

Microsoft Windows is the name given to a group of Operating Systems developed by Microsoft Corporation of **Bill Gates**.

OS	**Release Year**	**Comments**
Windows 3.x	1985	Windows 3.0, 3.1, 3.11, 3.111 and 3.2.
Windows 9.x Windows 95	 1995	MS Windows OS produced from 1995 to 2000. Windows 95, 98 and Windows ME (Millennium). Based on Windows 95 kernel and its MS-DOS. Had very few security mechanisms designed? Windows ME had minor enhancements of 98. Windows 95 replaced Windows 3.x in 1995.
Window NT	1993	Meant for users who work with large files and complex programs. Includes workstation and server versions. Windows NT 3.x versions include 3.5 and 3.51. Windows NT 4.0 proved most secure compared to earlier versions.
Windows 98	1998	It was the most popular Windows OS ever sold. (82% of OS market share).
Windows 2000	2000	Includes workstation and server versions For example, Windows 2000 Server. Offers a reasonable security when updated.

Windows XP		Replaced Windows 9.x family and 2000. Editions e.g. Starter, Home & Professional.
Windows Vista	2006	Initially code named **Longhorn**. Replaced Windows XP. Editions are Starter, Home Basic, Home Premium, Business, Enterprise and Ultimate.
Windows 7	2009	Incremental upgrade designed to work with Windows Vista compatible applications and hardware. Editions are Starter, Home Basic, Home Premium, Professional, Enterprise and Ultimate.

DOS (Disk Operating System)
-Has had many versions e.g. **MS DOS** (Microsoft DOS) used to be the main OS for IBM computers & **Apple DOS** for Macintosh computers.
-MS DOS 1.0 (1st edition in 1981) and MS DOS 7.0 (edition released in 1995).

Advantages of Windows over DOS

1. User friendliness-With GUI (Graphical User Interface).
2. Multiuser-Only Windows server versions are multiuser.
3. Multitasking-MS DOS runs only one program at a time.
4. Security enhancement-Higher versions from Windows 2000.

UNIX Systems
-1st developed by **Bell Telephone Research Labs** in 1969.
-Based on the model of **free distribution** and **open source** software development and distribution.
-Most installed in server machines in the history of PC manufacturing.
-There many versions e.g. Linux with also other **distributions** (versions).
Note:

- **Free distribution** means you can download for free use, without licensing.
- **Open source** means the source code is open to the public to contribute for improvement.

Compare Microsoft Software whose source code is highly restricted.

Linux Systems
The Linux kernel was first released in 1991 by **Linus Torvalds** (A student from Finland). It is a leading OS on servers and other mainframe computers and supercomputers. Linux is packaged in a format known as a **Linux distribution** for workstations and servers. Some popular Linux distributions are: Red Hat, SuSE, Ubuntu, Fedora & Mandriva.

How different is Linux from Windows?

1. Case sensitive
2. Server security
3. User accounts
4. Linux web hosting
5. Linux specific applications
6. Graphical user interfaces
7. Package manager
8. File structure of Linux
9. Command terminal

Case sensitive

-Linux is case sensitive and while Windows is not e.g. for Linux server: home.html, HOME.HTML and Home.html are all different files.

Server security

-Although many people consider Linux to be more secure than Windows, both operating systems are equally secure.

-Security really has more to do with the server setup and administrators running the server.

-All managed servers are maintained with the latest security and performance patches and configured for optimal security.

User accounts

-Linux built-in user accounts are: super user (**root**) and normal user.

-Super user (root) is the only administrator account used, while normal user is an ordinary account without privileges.

Linux web hosting

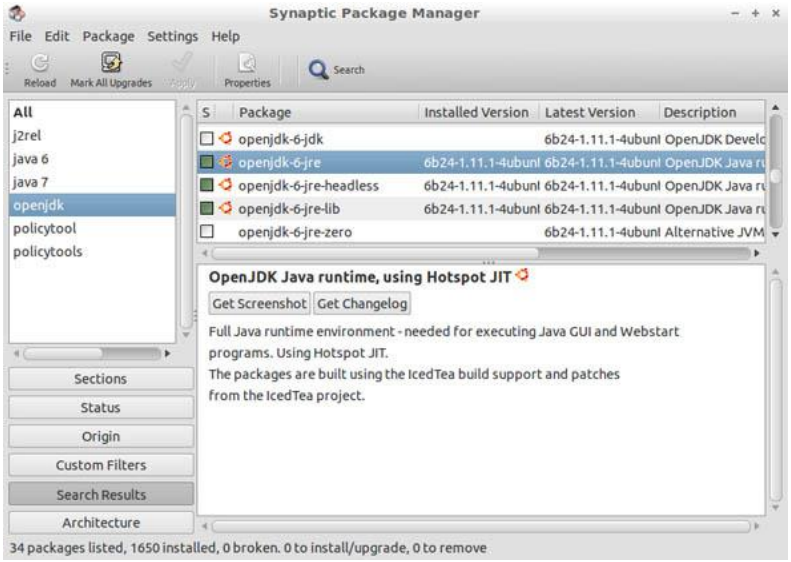
-Linux is the most popular OS for web servers, because it has more of the features web designers expect.

Graphical user interfaces

Interchangeable graphical user interfaces are:

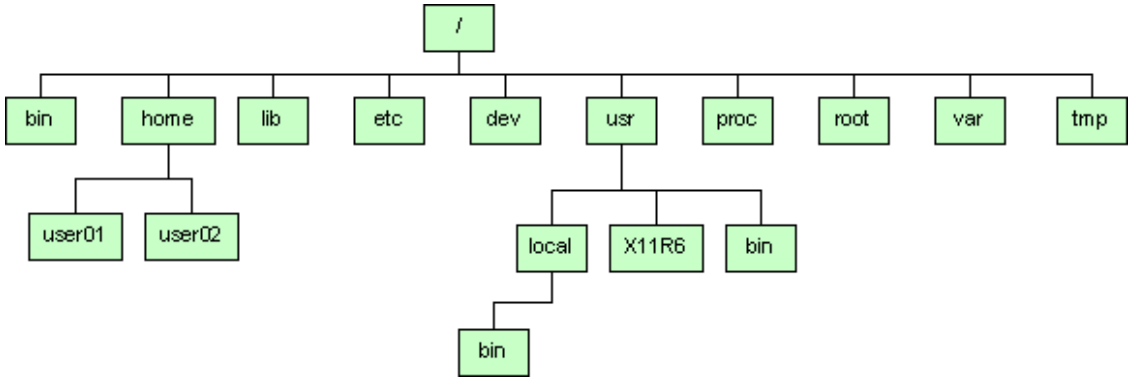
GNOME, KDE, Unity (more recent) & others (lesser-known varieties that all focus on different aspects)

Package manager



-This is the centre for browsing, installing and uninstalling programs.
-There is no need to visit the web browser websites (e.g. Firefox website).
-Just search the package manager's repositories and download the program

File structure of Linux



There is a **hierarchical directory structure** i.e. tree-like pattern of **directories** (called **folders** in Windows OS). Directories may contain files and also other directories.

Note:

- Linux system does not employ the concept of **drive letters**, like Windows OS. Instead there is a single tree.
- Different storage devices may contain different branches of the tree, but there is always a single tree.

Command terminal

```
beingjavaguys@beingjavaguys-pc:~/Documents/testdir$ ls
dir1  dir2  dir3
beingjavaguys@beingjavaguys-pc:~/Documents/testdir$ cd dir1
beingjavaguys@beingjavaguys-pc:~/Documents/testdir/dir1$ mkdir dirchild2
beingjavaguys@beingjavaguys-pc:~/Documents/testdir/dir1$ ls
dirchild2
beingjavaguys@beingjavaguys-pc:~/Documents/testdir/dir1$ pwd
/home/beingjavaguys/Documents/testdir/dir1
beingjavaguys@beingjavaguys-pc:~/Documents/testdir/dir1$ cd ../../
beingjavaguys@beingjavaguys-pc:~/Documents$ pwd
/home/beingjavaguys/Documents
beingjavaguys@beingjavaguys-pc:~/Documents$
```

-This is a program (also called a **terminal emulator**) that opens a window and lets you interact with the shell.
-Terminal emulators for most Linux distributions include:
gnome, konsole, xterm, rxvt, kvt, nxterm, and eterm.
-It is similar to Windows command prompt, but more frequently used eg opening config files in a text editor.
-Originally, Unix-like systems (eg Linux) had only CLIs (Command Line Interfaces), but later GUIs (Graphical User Interfaces) were added.
-GUIs are helpful for many tasks, but they are not good for all tasks. **Use any of the** window manager **(interchangeable GUIs) in Linux system to access the terminal emulator eg KDE has** console **and GNOME has** gnome-terminal**.**
-The **shell prompt** contains user name and machine name followed by a dollar sign eg [me@linuxbox me]$, where **me** is user name and **linuxbox me** is machine name.

1.9 Types of Application software

- Application software (**applications** or **software packages** or **packages**) allow end users to accomplish specific tasks.
- They are of 2 main types:
  1. Off-shelf (Commercially developed)
  2. Tailor-made (Tailored to customer)

Off-shelf (Commercially developed)
-Popular applications include:

- Word processors e.g. MS Word, Word Perfect, Word Star
- Spreadsheets e.g. MS Excel, Lotus-123, QuattroPro
- Databases e.g. MS Access, Oracle, SQL
- Accounting e.g. MS Money, QuickBooks, Pastel, Sage
- Presentation e.g. MS PowerPoint
- Desktop publishing e.g. MS Publisher, Page Maker

- Project management e.g. MS Project
- Website designing e.g. MS FrontPage Express, Macromedia Dreamweaver
- Computer Aided Design e.g. AutoCAD, Archie Card
- Animation e.g. Macromedia Director, Macromedia Flash, Adobe Image Ready, Adobe Premiere, Adobe After Effect
- Graphics e.g. Corel Draw, Adobe Photoshop, Adobe Illustrator
- Data analysis e.g. SPSS, EPIINFO

Note:

- Microsoft office suite includes MS Word, MS Excel, MS Access, MS PowerPoint etc.
- Windows OS also comes with simple programs which act as applications e.g. MS Paint, MS WordArt, MS ClipArt
- Tailor-made software may include e.g. Payroll Software for given organization, Software for given Bank & Software for Hotel Reservation.